Daksh Gupta

Blacksburg, VA | mailtodaksh@vt.edu | Portfolio: advanceuser2005.github.io/ | GitHub: github.com/advanceuser2005

EDUCATION

Virginia Tech College of Engineering **Expected May 2028**
Bachelor of Computer Engineering - *Controls, Robotics and Autonomy focus*
Relevant Coursework: Embedded Systems, Circuits and Devices, Data Structures and Algorithms, Computational Engineering

SKILLS

Programming: C++, Python, C, MATLAB, Verilog
Robotics: ROS 2 Humble/Jazzy, Nav2, SLAM, Path Planning, Autonomous Navigation, Gazebo, Isaac Sim
Software/Systems: Data Structures, Algorithms, Object-Oriented Programming, Linux, Ubuntu, Docker, Git/GitHub, CMake
Hardware/Embedded: Jetson Orin Nano, Teensy 4.1, Arduino, RealSense, LiDAR
EE/Design Tools: Fusion 360, KiCad/Altium, Quartus, Questa

ROBOTICS SOFTWARE EXPERIENCE

Robotics Software Engineer, VT Competitive Robotics Organization **September 2024 - May 2026**
- Developed ROS 2 robotics software for autonomous competition robots, including motion control, navigation, simulation, and hardware integration.
- Tuned PID-based motion-control nodes to improve drive stability, path tracking, and robot responsiveness during testing.
- Debugged odometry, TF transforms, localization, and navigation-stack issues across simulation and hardware integration.
- Built a custom local telemetry server to stream Isaac ROS nvblox data from the robot/simulation stack for real-time visualization, debugging, and mapping.
- Coordinated software tasks across navigation, simulation, perception, and embedded interfaces for a multi-member robotics team.

SELECTED PROJECTS

Autonomous Warehouse Robot + Fleet Telemetry Platform **July 2026 - Present**
Independent Project
- Building a ROS 2 warehouse robot simulation with task execution, navigation, obstacle avoidance, and robot health monitoring.
- Developing custom ROS 2 nodes/services for mission control, robot state reporting, and command handling.
- Designing telemetry architecture to publish robot pose, battery state, task status, and error logs for AWS-based fleet monitoring.
- Building the project with Linux, Docker, logging, and documentation to emphasize production-style robotics software practices.

Mini Tesla - Autonomous Mecanum EV Robot Platform **April 2026 - Present**
VT AMP Lab Project
- Designing and prototyping a mecanum-drive robotic EV platform with embedded motor control, regenerative braking architecture, and ROS 2 autonomy.
- Developing embedded control firmware for encoder feedback, PID velocity control, braking logic, and Teensy–Jetson communication.
- Building simulation-first workflow in Gazebo to validate robot behavior, navigation logic, and emergency braking before hardware deployment.

Lunar ResQ - Autonomous Rescue and Communication Robot and Drone **September 2025 - May 2026**
VT CRO IEEE SoutheastCon Hardware Competition 2026
- Developed ROS 2 Jazzy navigation and SLAM workflows for an autonomous ground rover using depth camera and IMU-based sensing.
- Built behavior-tree mission logic for coordinated rover/drone task execution within 3-minute competition constraints.
- Simulated crater traversal, path planning, and cooperative robot behavior in Gazebo and Isaac Sim.
- Worked on a rocker-bogie drivetrain requiring custom ROS 2 control integration for non-standard robot motion and terrain traversal.

Mining Mayhem - Autonomous Object Sorting Robot **September 2024 - March 2025**
VT CRO IEEE SoutheastCon Hardware Competition 2025
- Built ROS-based mecanum-drive robot software for autonomous navigation, mapping, and object-sorting tasks.
- Used AprilTags and wheel odometry for localization, target detection, and alignment.

LEADERSHIP

Founder, MoonLit (Startup) **Jun 2023 - June 2024**
- Designed an ESP32–MQTT IoT platform for plant monitoring, automation, real-time visualization, and remote management.
- Built a React Native app prototype for remote device monitoring and control.

President, Robotics Club & Cultural Society **May 2021 - May 2023**
- Taught Python to 120+ students and organized 10+ STEM/cultural events promoting robotics and programming.
- Led robotics outreach initiatives reaching 150+ peers across technical and cultural programs.